SUBMERSIONS AND GEODESICS

By Barrett O'Neill

1. Introduction. For Riemannian manifolds M and B, a *submersion* $\pi : M \to B$ is a mapping onto B which has maximal rank and preserves lengths of horizontal tangent vectors to M [4]. (A tangent vector to M at p is *horizontal* if orthogonal to the *fiber* $\pi^{-1}\pi(p)$ through p, *vertical* if tangent to the fiber.) Submersions occur frequently in Riemannian geometry; many examples are considered in [3], [6], notably the projection $G \to G/H$ of a Riemannian homogeneous space. A submersion $\pi : M \to B$ is described infinitesimally by tensors T and A on M introduced in [6], where the relations between the curvatures of M, B, and the fibers $\pi^{-1}(b)$ are expressed in terms of these tensors.

Our aim here is to compare the geodesics of M and B, and to show how conjugacy and index on a geodesic in B derive from the geometry of M and the fibers. When M is complete, any geodesic segment β in B is the projection $\pi \circ \gamma$ of a horizontal geodesic segment γ in M. Our main result is a formula (Theorem 3) relating the index form on γ, with fibers as endmanifolds, to the fixed endpoint index form on $\beta = \pi \circ \gamma$. As a consequence we can identify conjugacy and index on β with conjugacy and index on γ-with fibers as end-manifolds at either one or both endpoints of γ. When only fixed endpoint conjugacy on γ is known we still obtain information about conjugacy on β (Theorem 5). Roughly speaking, if a conjugate point on γ does not project to a conjugate point on β (order of conjugacy, at least, need not be preserved) it is nevertheless responsible for a conjugate point occurring earlier on β. Finally we look at a special case which suggests how detailed information about M and the tensor A can be used to locate conjugate points in B.

For the curvature relations mentioned above we refer to [6], but we now summarize briefly the basic properties of the tensors T and A. Let $\mathcal{H}$ and $\mathcal{V}$ denote the projections of each tangent space of M onto its (complementary) subspaces of horizontal and vertical vectors. Then for arbitrary vector fields E and F on M, $T_E F = \mathcal{H}\nabla_{\mathcal{V}E}(\mathcal{V}F) + \mathcal{V}\nabla_{\mathcal{V}E}(\mathcal{H}F)$. Thus T is one formulation of the second fundamental form of all fibers. For $A_E F$, simply exchange $\mathcal{H}$ and $\mathcal{V}$ in the formula for $T_E F$. Subsequent computations make frequent use of the following properties of T and A: (1) T_E and A_E are, at each point, skew-symmetric linear operators on the tangent spaces of M; each sends horizontal vectors to vertical, and vertical to horizontal; (2) T is vertical, A horizontal, that is, $T_E = T_{\mathcal{V}E}$ and $A_E = A_{\mathcal{H}E}$; (3) for vertical vector fields, $T_V W = T_W V$; for horizontal vector fields, $A_X Y = \frac{1}{2}\mathcal{V}[X, Y] = -A_Y X$. This last fact, proved in [6], shows that A is the integrability tensor of the horizontal distribution on M.

Received June 6, 1966. This work was supported by a National Science Foundation grant.

2. Geodesics. If $\pi : M \to B$ is a submersion, then to relate the geodesics (and later Jacobi fields) of M to those of B we need to compare the covariant derivatives of M and B as applied to vector fields on curves. Fix the following notation: if E is a vector field on a curve α in M, then H is the horizontal part $\mathcal{H}E$ of E, V the vertical part $\mathcal{V}E$, and E_* is the vector field $\pi_*(E) = \pi_*(H)$ on the curve $\pi \circ \alpha$ in B. The covariant derivative of a vector field on a curve is denoted by a prime; thus E_*' is a vector field on $\pi \circ \alpha$. But we use the same notation for its horizontal lift to α, so E_*' also denotes the horizontal vector field on α that projects to the covariant derivative of $\pi_*(E)$.

THEOREM 1. *Let $\pi : M \to B$ be a submersion, and let $E = H + V$ be a vector field on a curve α in M. Then*

$$\mathcal{H}(E') = E_*' + A_H(\mathcal{V}\alpha') + A_{\mathcal{H}\alpha'}(V) + T_{\mathcal{V}\alpha'}(V)$$
$$\mathcal{V}(E') = A_{\mathcal{H}\alpha'}(H) + T_{\mathcal{V}\alpha'}(H) + \mathcal{V}(V').$$

Proof. We work in a neighborhood of an arbitrary point $\alpha(t)$ of the curve, and assume $\alpha'(t) \neq 0$. (The case $\alpha'(t) = 0$ can be handled as in the last paragraph of this proof.) Then the vector fields $\mathcal{H}\alpha'$ and $H = \mathcal{H}E$ on α have extensions to horizontal vector fields X and H on a neighborhood of $\alpha(t)$, and $\mathcal{V}\alpha'$ and $V = \mathcal{V}E$ have vertical extensions U and V. On the curve α,

$$\mathcal{H}(E') = \mathcal{H}(H') + \mathcal{H}(V') \quad \text{and} \quad \mathcal{V}(E') = \mathcal{V}(H') + \mathcal{V}(V'),$$

so the following three assertions are needed:

$$\mathcal{H}(V') = A_{\mathcal{H}\alpha'}(V) + T_{\mathcal{V}\alpha'}(V) \tag{1}$$

The vector field $\mathcal{H}(V')$ is the restriction to α of $\mathcal{H}\nabla_{X+U}(V) = \mathcal{H}\nabla_X V + \mathcal{H}\nabla_U V$. But by the definitions of T and A given earlier, the latter is $A_X V + T_U V$. Similarly we have

$$\mathcal{V}(H') = A_{\mathcal{H}\alpha'}(H) + T_{\mathcal{V}\alpha'}(H). \tag{2}$$

To prove

$$\mathcal{H}(H') = E_*' + A_H(\mathcal{V}\alpha'), \tag{3}$$

we first assume that $\mathcal{H}\alpha'(t) \neq 0$. Thus, near t, the curve $\pi \circ \alpha$ is regular, and it follows easily that the extension H used above can be chosen to be *basic*, that is, horizontal and π-related to a vector field on B. Now $\mathcal{H}H'$ is the restriction to α of $\mathcal{H}\nabla_X H + \mathcal{H}\nabla_U H$. Since H is basic and U is vertical, the bracket $[H, U]$ is vertical, hence the first structural equation implies $\mathcal{H}\nabla_U H = \mathcal{H}\nabla_H U = A_H U$. By Lemma 1 of [6], $\mathcal{H}\nabla_X H$ is the horizontal lift of the covariant derivative of $\pi_*(H) = \pi_*(E) = E_*$ with respect to $\pi_*(X)$. But $\pi_*(X)$ is an extension of $(\pi \circ \alpha)'$, hence the restriction of $\mathcal{H}\nabla_X H$ to α is E_*'.

This argument fails when $\mathcal{H}\alpha'(t) = 0$. In this case, let U be merely a vertical vector field whose value at $\alpha(t)$ is $\alpha'(t)$. Then $\mathcal{H}H'(t)$ is the value at $\alpha(t)$ of the single vector field $\mathcal{H}\nabla_U H$. It is generally not possible to choose the extension

H to be basic, so let $B_1, \cdots, B_b$ be the horizontal lift of a frame field on B near $\pi(\alpha(t))$, and write $H = \Sigma h_i B_i$. Then

$$\mathcal{H}\nabla_U H = \Sigma U(h_i)B_i + \Sigma h_i \mathcal{H}\nabla_U(B_i)$$

Since the B_i's are basic, we have $\mathcal{H}\nabla_U B_i = \mathcal{H}\nabla_{B_i}(U) = A_{B_i}(U)$. Hence $\Sigma h_i \mathcal{H}\nabla_U(B_i) = A_H U$, which at $\alpha(t)$ gives the required A term. It is an exercise in notation to check that the value of $\Sigma U(h_i)B_i$ at $\alpha(t)$ is $E'_*(t)$.

Write α'' for the acceleration of α, that is, the covariant derivative of α'. Since $A_Y Y = 0$, setting $E = \alpha'$ in the theorem yields:

COROLLARY 1. *Let α be a curve in M with $X = \mathcal{H}\alpha'$ and $U = \mathcal{V}\alpha'$. Then*

$$\mathcal{H}(\alpha'') = \alpha''_* + 2A_X U + T_U U$$
$$\mathcal{V}(\alpha'') = T_U X + \mathcal{V}(U')$$

where α''_* is the horizontal lift to α of the acceleration of $\pi \circ \alpha$ in B.

Setting $\mathcal{H}\alpha'' = \mathcal{V}\alpha'' = 0$ thus gives a pair of equations necessary and sufficient for α to be a geodesic of M. If α is a geodesic, then $\pi \circ \alpha$ is a geodesic of B if and only if $2A_X U + T_U U = 0$. In particular, if α is a *horizontal* geodesic ($U = \mathcal{V}\alpha' = 0$), then $\pi \circ \alpha$ is geodesic. (It is well known from special cases of submersions that arbitrary geodesics of M need not project to geodesics.)

The following fundamental result is due to R. Hermann. His proof [5], [4] uses length-minimizing properties of geodesics. We derive an infinitesimal proof from Theorem 1.

COROLLARY 2. *Let $\pi : M \to B$ be a submersion. If γ is a geodesic of M that is horizontal at some one point, then γ is always horizontal (hence $\pi \circ \gamma$ is a geodesic of B).*

Proof. For $E = \gamma'$, assertion (1) in the proof of Theorem 1 gives $\mathcal{H}U' = A_X U + T_U U$, where as usual $X = \mathcal{H}\gamma'$ and $U = \mathcal{V}\gamma'$. But by Corollary 1, $\mathcal{V}U' = -T_U X$. Thus $U' = A_X U + T_U U - T_U X$. Obviously the zero vector field is a solution of this differential equation; by hypothesis U is somewhere zero, so by uniqueness we conclude $U = 0$, that is, γ is horizontal.

The corollary may be interpreted as saying that the horizontal distribution $\mathcal{H}$—though integrable if and only if $A = 0$—is always "totally geodesic". When M is complete, it implies that every geodesic in B is the projection of a horizontal geodesic in M, thus permitting the geodesic structure of B to be studied in M.

3. Comparison of index forms. From now on, γ will denote a horizontal geodesic in M, where $\pi : M \to B$ is a submersion. We shall compare the index form for a segment of γ (with suitable boundary conditions) and the index form for its projection in B. The theorem below, which analyzes the second covariant derivative of a vector field E on γ, gives this comparison in infinitesimal form. We write $X = \gamma'$ (horizontal), and as usual H and V are the horizontal and vertical parts of E.

DEFINITION. If $E = H + V$ is a vector field on γ, then

$$D(E) = \mathcal{V}(V') - T_V X + 2A_X H$$

is the *derived vector field* of E.

Evidently $D(E)$ is a vertical vector field on γ; we shall see in a moment that it gives the end point terms in the index form.

THEOREM 2. *Let E be a vector field on a horizontal geodesic γ in M. Then*

$$\mathcal{H}(E'' - R_{EX}X) = (E''_* - R^*_{E_*X}X) + 2A_X D$$
$$\mathcal{V}(E'' - R_{EX}X) = \mathcal{V}(D') + T_D X$$

where $D = D(E)$ is the derived vector field of E, and R and R^* are the curvature tensors of M and B.

(The term $E''_* - R^*_{E_*X}X$ denotes the horizontal lift to γ of the "Jacobi expression" for $E_* = \pi_*(E)$ on $\pi \circ \gamma$.)

Proof. We write either E' or $\nabla_X E$ for covariant derivatives of vector fields on γ. Since $X = \gamma'$ is horizontal,

$$H' = \mathcal{H}\nabla_X H + A_X H, \qquad V' = A_X V + \mathcal{V}\nabla_X V.$$

These formulas can now be applied to the horizontal and vertical parts of H' and V'. Thus $E'' = H'' + V''$ is the sum of the following terms

$$\begin{array}{ll} \mathcal{H}\nabla_X(\mathcal{H}\nabla_X H) + A_X(\mathcal{H}\nabla_X H) & A_X A_X H + \mathcal{V}\nabla_X(A_X H) \\ \mathcal{H}\nabla_X(A_X V) + A_X A_X V & A_X(\mathcal{V}\nabla_X V) + \mathcal{V}\nabla_X(\mathcal{V}\nabla_X V). \end{array}$$

According to Theorem 1, $\mathcal{H}\nabla_X H$ is (the horizontal lift of) E'_*, hence $\mathcal{H}\nabla_X(\mathcal{H}\nabla_X H)$ is E''_*. Thus we have

(1) $\quad \mathcal{H}E'' = E''_* + A_X A_X H + \mathcal{H}\nabla_X(A_X V) + A_X(\mathcal{V}\nabla_X V)$

(2) $\quad \mathcal{V}E'' = A_X(\mathcal{H}\nabla_X H) + \mathcal{V}\nabla_X(A_X H) + A_X A_X V + \mathcal{V}\nabla_X(\mathcal{V}\nabla_X V).$

To express $R_{EX}X = R_{HX}X + R_{VX}X$ in similar terms we use results from [6]. For an arbitrary horizontal vector field Y on γ, Theorem 2 of [6] gives $\langle R_{HX}X, Y\rangle = \langle R^*_{E_*X}X, Y\rangle - 3\langle A_H X, A_X Y\rangle$. But $A_H X = -A_X H$, and A_X is skew-symmetric, hence

(3) $$\mathcal{H}R_{HX}X = R^*_{E_*X}X - 3A_X A_X H.$$

By the same theorem, $\langle R_{VX}X, Y\rangle = \langle(\nabla_X A)_X V, Y\rangle - 2\langle T_V X, A_X Y\rangle$. But $(\nabla_X A)_X V = \nabla_X(A_X V) - A_X(\nabla_X V)$, since $\nabla_X X = \gamma'' = 0$. Using $\mathcal{H}A_X = A_X\mathcal{V}$, we obtain

(4) $$\mathcal{H}R_{VX}X = \mathcal{H}\nabla_X(A_X V) - A_X(\mathcal{V}\nabla_X V) + 2A_X T_V X.$$

Subtracting (3) and (4) from (1) we get the required formula for $\mathcal{H}(E'' - R_{EX}X)$.

In a similar way the curvature formulas in Theorems 2 and 3 of [6] give

$$\mathcal{V}R_{HX}X = -\mathcal{V}\nabla_X(A_XH) + A_X(\mathcal{H}\nabla_XH) - 2T_{A_XH}(X)$$
$$\mathcal{V}R_{VX}X = \mathcal{V}\nabla_X(T_VX) - T_{\mathcal{V}\nabla_XV}(X) + T_{T_VX}(X) + A_XA_XV.$$

Subtracting these two equations from (2) then yields the required formula for $\mathcal{V}(E'' - R_{EX}X)$.

This result gives precise information on the relation of Jacobi fields on γ to those on $\pi \circ \gamma$; for example, it shows that a Jacobi field E on γ projects to a Jacobi field on $\pi \circ \gamma$ if and only if $A_X(D(E)) = 0$.

LEMMA 1. *Let γ be a horizontal geodesic in M. Given a vector field P on $\pi \circ \gamma$ and a vertical vector v at, say, $\gamma(0)$, there exists a unique vector field E on γ such that* (1) $\pi_*(E) = P$, (2) $D(E) = 0$, *and* (3) $E(0) = v$. *Furthermore, E is a Jacobi field if and only if P is.*

Proof. If E is such a vector field, then by (1) $H = \mathcal{H}E$ is uniquely determined as the horizontal lift of P to γ. By (2), $V = \mathcal{V}E$ satisfies $\mathcal{V}V' = T_VX - 2A_XH$. Since $\mathcal{H}V' = A_XV$, we have

$$V' = A_XV + T_VX - 2A_XH. \tag{*}$$

Together with (3) this uniquely determines V and hence $E = H + V$.

To reverse this argument and show that E exists it suffices to verify that if V is any solution of (*) with $V(0)$ vertical, then V is everywhere vertical. (To prove this deduce from (*) that $(\mathcal{H}V)' = A_X(\mathcal{H}V)$.) The final assertion of the lemma follows from the preceding theorem.

Now suppose that $\gamma : [a, b] \to M$ is a horizontal geodesic segment, and let F_t be the fiber $\pi^{-1}\pi\gamma(t)$ through $\gamma(t)$. For the geodesic segment $\pi \circ \gamma$ in B, conjugacy and index derive from the linear space $\mathcal{B}$ consisting of all piecewise differentiable (C^∞) vector fields on $\pi \circ \gamma$ that are zero at a and b, and everywhere orthogonal to the curve. A piecewise differentiable vector field E on will have projection E_* in $\mathcal{B}$ if and only if E is orthogonal to γ and vertical at a and b. Such vector fields constitute the linear space $\mathcal{E}^{vv}$ appropriate to the study of γ as a geodesic segment joining the fibers F_a and F_b through its end points. (All first variations of γ are zero since, being horizontal, γ is orthogonal to all fibers.) The index form I on $\mathcal{E}^{vv}$ is given [2] by

$$I(E, F) = -\int_a^b \langle E'' - R_{EX}X, F\rangle\, dt + \langle E' - S_X(E), F\rangle]_a^b + \Sigma_t\langle \Delta E'(t), F(t)\rangle$$

where $X = \gamma'$, S is the classical second fundamental form, and $\Delta E'(t) = E'(t^-) - E'(t^+)$.

We rewrite the non-integrated terms using the tensor T and the derived vector field $D = D(E)$ of E. In general, for v vertical (tangent to fiber) and x horizontal (orthogonal to fiber), $S_xv = T_vx$. Writing $E = H + V$ as usual, we have $\langle S_XE, F\rangle]_a^b = \langle T_VX, F\rangle]_a^b$, since $E = V$ at a and b. Also $\langle E', F\rangle]_a^b =$

$\langle \mathcal{V}V', F\rangle]_a^b$, since $\mathcal{V}E' = A_XH + \mathcal{V}V'$. At a non-differentiable point t for E, we get $\Delta E'(t) = \Delta(\mathcal{V}V') + \Delta(\mathcal{H}H') = \Delta D(t) + \Delta E_*'(t)$, since H and V are continuous at t. Thus the formula above for the index form becomes

$$I(E, F) = -\int_a^b \langle E'' - R_{EX}X, F\rangle\, dt + \langle D, F\rangle]_a^b$$
$$+ \Sigma_t\langle \Delta D(t), F(t)\rangle + \Sigma_t\langle E_*'(t), F_*(t)\rangle.$$

THEOREM 3. *Let $\gamma : [a, b] \to M$ be a horizontal geodesic segment where $\pi : M \to B$ is a submersion. If I is the index form on $\mathcal{E}^{vv}$ and I_B is the index form on $\mathcal{B}$, then*

$$I(E, F) = I_B(E_*, F_*) + \int_a^b \langle D(E), D(F)\rangle\, dt$$

where $D(E)$ and $D(F)$ are the derived vector fields of E and F on γ.

Proof. It follows from Theorem 2 that

$$\langle E'' - R_{EX}X, F\rangle = \langle E_*'' - R^*_{E_*X}X, F_*\rangle + 2\langle A_XD, F\rangle + \langle \mathcal{V}D' + T_DX, F\rangle$$

where $D = D(E)$. Since D is vertical, $\langle \mathcal{V}D', F\rangle = \langle D', \mathcal{V}F\rangle = \langle D, F\rangle' - \langle D, (\mathcal{V}F)'\rangle$. Using the skew-symmetry of A_X and T_D we get

$$\langle E'' - R_{EX}X, F\rangle = \langle E_*'' - R^*_{E_*X}X, F_*\rangle + \langle D(E), F\rangle' - \langle D(E), D(F)\rangle$$

which integrates to the required formula. (There are no boundary terms in I_B ; we consider only the fixed endpoint problem for $\pi \circ \gamma$.)

4. Conjugacy and index. As before, $\gamma : [a, b] \to M$ denotes a horizontal geodesic segment. Let $\mathcal{J}$ be the linear space of all Jacobi fields on γ orthogonal to γ, and consider its subspaces: $\mathcal{J}_\delta = \{E \mid D(E) = 0\}$, $\mathcal{J}_v = \{E \mid E \text{ is vertical}\}$, and $\mathcal{J}^{vv} = \{E \mid E(a) \text{ and } E(b) \text{ are vertical}\}$. In the last case, replacement of either "v" by a zero means vanishing instead of verticalness; and the notations may be combined in an obvious way.

LEMMA 2. *The space $\mathcal{J}_{\delta v}$ (vertical Jacobi fields with $D(E) = 0$) consists of all solutions of $E' = A_XE + T_EX$ on γ such that $E(a)$ is vertical. In particular, $\mathcal{J}_{\delta v}$ has dimension k, where $k = \dim M - \dim B$ is the common dimension of all fibers.*

Proof. If V is merely a vertical vector field on γ, then $D(V) = 0$ becomes $\mathcal{V}V' = T_VX$. But $\mathcal{H}V' = A_XV$, so $V' = A_XV + T_VX$.

Conversely, substituting $E = H + V$ in the differential equation gives $H' + V' = A_XH + A_XV + T_VX$ (since $T_H = 0$). Applying $\mathcal{H}$ to this equation yields $\mathcal{H}H' = 0$, hence $H' = A_XH$. By hypothesis $H(a) = 0$, so $H = 0$, that is, $E = V$. The equation then reduces to $\mathcal{V}V' = T_VX$, which is the same as $D(E) = 0$. Finally, by Theorem 2, $E = V$ is a Jacobi field and is orthogonal to γ, so E is in $\mathcal{J}_{\delta v}$.

If E is a Jacobi field on γ with derived vector field D, then by Theorem 2, $\mathcal{V}D' + T_D X = 0$, hence $D' = A_X D - T_D X$. So if D vanishes at one point, then $D = 0$, that is, E is in $\mathcal{J}_\delta$.

The nullspace of the index form I on $\mathcal{E}^{vv}$ consists of all E in $\mathcal{J}^{vv}$ such that $D(a) = D(b) = 0$ (see [2; 221, Theorem 1]). Since $D(a) = 0$ alone implies $D = 0$, this nullspace is $\mathcal{J}_\delta^{vv}$. Similarly, the nullspace of I on $\mathcal{E}^{v0}[\mathcal{E}^{0v}]$ is $\mathcal{J}_\delta^{v0}[\mathcal{J}_\delta^{0v}]$. The dimensions of the latter thus give the order of $\gamma(b)[\gamma(a)]$ as a focal point of the fiber $F_a[F_b]$ along γ. In the two endmanifold case, however, Ambrose [1] stresses that the order of conjugacy of F_a and F_b along γ should *not* be defined to be the nullity of the index form—in our case the dimension of $\mathcal{J}_\delta^{vv}$. For the moment we define this order to be dim $(\mathcal{J}_\delta^{vv}/\mathcal{J}_{\delta v}) = \dim(\mathcal{J}_\delta^{vv}) - k$, where k is the fiber dimension. After Corollary 3 we relate this to Ambrose' general definition.

THEOREM 4. *Let $\pi : M \to B$ be a submersion, $\gamma : [a, b] \to M$ a horizontal geodesic segment. Then the following integers are equal:*

1. *The order of $\gamma(b)$ as a focal point of the fiber F_a along γ,*
2. *The order of $\gamma(a)$ as a focal point of the fiber F_b along γ,*
3. *The order of conjugacy of F_a and F_b along γ,*
4. *The order of conjugacy of the end points of $\pi \circ \gamma$ along $\pi \circ \gamma$.*

Proof. By Theorem 2, π_* carries $\mathcal{J}_\delta^{vv}$ (hence $\mathcal{J}_\delta^{0v}$) into $\mathcal{J}_B^{00}$ (Jacobi fields on $\pi \circ \gamma$ zero at end points and orthogonal to $\pi \circ \gamma$). By Lemma 1, π_* carries $\mathcal{J}_\delta^{0v}$ (hence $\mathcal{J}_\delta^{vv}$) *onto* $\mathcal{J}_B^{00}$. The kernel of π_* on $\mathcal{J}_\delta^{vv}$ is $\mathcal{J}_{\delta v}$, but π_* is one-one on $\mathcal{J}_\delta^{0v}$, since Lemma 2 shows that an element of $\mathcal{J}_{\delta v}$ zero at one point is identically zero. Evidently $\mathcal{J}_\delta^{0v}$ may be replaced throughout by $\mathcal{J}_\delta^{v0}$, so we conclude that $\dim \mathcal{J}_\delta^{v0} = \dim \mathcal{J}_\delta^{0v} = \dim(\mathcal{J}_\delta^{vv}/\mathcal{J}_{\delta v}) = \dim \mathcal{J}_B^{00}$.

In particular the orders in 1, 2, and 3 cannot exceed dim $B - 1$.

COROLLARY 3. *The indexes of γ as a geodesic segment from F_a to $\gamma(b)$, or $\gamma(a)$ to F_b, or F_a to F_b are all the same as the (fixed endpoint) index of $\pi \circ \gamma$.*

This asserts the equality of the index of I on $\mathcal{E}^{v0}$, on $\mathcal{E}^{0v}$, on $\mathcal{E}^{vv}$, and the index of I_B on $\mathcal{B}$. Technically it is a consequence of the preceding theorem and suitable formulations of the Index Theorem; however the following direct proof (particularly in the two endmanifold case) is much simpler:

Let $\mathcal{W}$ be a subspace of $\mathcal{E}^{vv}$ on which the index form I is negative definite. Then π_* is one-one on $\mathcal{W}$, and I_B is negative definite on $\pi_*\mathcal{W}$. In fact, if π_* carries $W \in \mathcal{W}$ to zero, then by Theorem 3, $0 \geq I(W, W) = \int_a^b ||D(W)||^2 \, dt \geq 0$. But for $W \in \mathcal{W}$, $I(W, W) = 0$ implies $W = 0$. The negative definite assertion then follows from the same theorem. Thus the first three indexes are no greater than index I_B.

For the reverse inequality, let $\mathcal{P}$ be a subspace of $\mathcal{B}$ on which I_B is negative definite. Since $\mathcal{E}^{0v} \subset \mathcal{E}^{vv}$, it suffices to consider I on $\mathcal{E}^{0v}$. Let $\mathcal{W}$ be the subspace $\{E \in \mathcal{E}^{0v} \mid E_* \in \mathcal{P} \text{ and } D(E) = 0\}$. By Lemma 1, π_* carries $\mathcal{W}$ onto $\mathcal{P}$. We claim that I is negative definite on $\mathcal{W}$. Using Theorem 3 we have $I(W, W) =$

$I_B(W_*, W_*) \leq 0$ for $W \in \mathcal{W}$. If equality holds, then $W_* = 0$, and the proof of Lemma 2 shows that $W \in \mathcal{J}_{\delta v}$; hence $W(a) = 0$ implies $W = 0$.

We now examine the two endmanifold case. If $a < t \leq b$, then on the segment $\gamma_t = \gamma \mid [a, t]$, the various spaces of vector fields are denoted by the scheme $\mathcal{E}^{vv}(t)$ (defined only on $[a, t]$, vertical at a and t). The last two results together with the fixed endpoint form of the Morse Index Theorem then show that

$$\text{Index } I \text{ on } \mathcal{E}^{vv} = \sum_{a<t<b} \dim\left(\mathcal{J}_\delta^{vv}(t)/\mathcal{J}_{\delta v}(t)\right).$$

This is Ambrose' formulation of the Index Theorem for two endmanifolds—in our very special case. (The convexity term is zero.) In Ambrose' general definition of conjugacy at $\gamma(t)$, the boundary conditions supplied by the endmanifold at $\gamma(b)$ are moved by infinitesimal means to $\gamma(t)$; in our case these conditions are automatically supplied by the fiber F_t through $\gamma(t)$. Then it is necessary to divide out the "permanent nullity"—in our case, $\mathcal{J}_{\delta v}(t)$—since as the Index Theorem shows, this subspace of the nullspace $\mathcal{J}_\delta^{vv}(t)$ makes no contribution to index as t increases from a to b. More intuitively: the elements of $\mathcal{J}_{\delta v}(t)$ are associated vector fields of rectangles whose longitudinal curves are, like γ_t, horizontal geodesic lifts of $\pi \circ \gamma_t$, and thus have the same length as γ_t itself. Their character is unchanged as t increases to b; they do not lead to routes from F_a to F_b shorter than γ.

So far we have had at least one fiber as endmanifold in dealing with the horizontal geodesic segment γ. Now we consider the fixed endpoint case, writing $\omega = \omega(\gamma)$ for the order of conjugacy of the endpoints of γ along γ, and $i(\gamma)$ for the corresponding (fixed endpoint) index. The order $\omega(\gamma)$ bears no obvious relation to $\omega(\pi \circ \gamma)$—neither inequality holds—however we shall find an inequality for indexes. We have $\omega = \omega(\gamma) = \dim \mathcal{J}^{00}$; define $\omega_\delta = \dim \mathcal{J}_\delta^{00}$, and $\omega_n = \dim(\mathcal{J}^{00}/\mathcal{J}_\delta^{00})$. Thus $\omega = \omega_\delta + \omega_n$, and we assert that $\omega_\delta \leq \dim B - 1$ and $\omega_n \leq k$ (fiber dimension). It suffices to show that $\mathcal{J}_\delta^{00} = \{E \in \mathcal{J}^{00} \mid E'(a) \text{ is horizontal}\}$. But if $E = H + V$ is a Jacobi field on γ such that $E(a) = 0$, then $\mathcal{V}E'(a) = \mathcal{V}V'(a) = D(a)$, and (as noted earlier) if the derived vector field D is zero at one point it is identically zero.

In particular, $\omega_n > 0$ if the order of conjugacy ω is at least as large as the dimension of B.

THEOREM 5. *Let γ be a horizontal geodesic segment in M, where $\pi : M \to B$ is a submersion. Then (with notation as above) $i(\pi \circ \gamma) \geq i(\gamma) + \omega_n(\gamma)$.*

Proof. Let $\mathcal{W}$ be a subspace of $\mathcal{E}^{00}$ on which the index form I is negative definite; we can assume $\dim \mathcal{W} = i(\gamma)$. Let $\mathcal{C}$ be a subspace of $\mathcal{J}^{00}$ complementary to $\mathcal{J}_\delta^{00}$, so $\dim \mathcal{C} = \omega_n$. It suffices to prove that (1) π_* is one-one on the (direct) sum $\mathcal{C} + \mathcal{W}$, and (2) I_B is negative definite on $\pi_*(\mathcal{C} + \mathcal{W})$.

We write $I(Y) = I(Y, Y)$. Since $\mathcal{C}$ is in the nullspace of I on $\mathcal{E}^{00}$, we have $I(E + W) = I(W)$ for $E \in \mathcal{C}$, $W \in \mathcal{W}$. Thus it follows from Theorem 3 that

I_B is at least negative semi-definite on $\pi_*(\mathcal{C} + \mathcal{W})$. Then assertions (1) and (2) will follow from (3) $I_B(E_* + W_*) = 0$ implies $E = W = 0$, for $E \in \mathcal{C}$, $W \in \mathcal{W}$. Using Theorem 3 we have $0 \geq I(W) = I(E + W) = \int_a^b ||D(E + W)||^2 \, dt \geq 0$. But by definition of $\mathcal{W}$, $I(W) = 0$ implies $W = 0$. Then we are left with $\int_a^b ||D(E)||^2 \, dt = 0$, so $D(E) = 0$. Hence by definition of $\mathcal{C}$ we have $E = 0$.

Because of Theorem 4 we can use this inequality (in favorable cases) to deduce the existence of focal points of F_a (or two equivalent possibilities from Theorem 4) along γ from information about fixed endpoint conjugacy only. Explicitly, set $\omega_v(\gamma) = \dim (\mathcal{J}_\delta^{v0}/\mathcal{J}_\delta^{00}) \leq k$. Then the order of $\gamma(b)$ as a focal point of F_a is $\omega_\delta + \omega_v$. Here ω_v is the strictly focal part; it counts dimensions for the Jacobi fields which contribute to focal order but not conjugate order. If $\gamma_t = \gamma \mid [a, t]$ for $a < t \leq b$, then the preceding result, together with the Morse Index Theorem, shows that $\Sigma_{t<b}\omega_v(\gamma_t) \geq \Sigma_{t\leq b}\omega_n(\gamma_t)$; that is, the total number (counting orders) of strict focal points *before* b is no less than the total "excess conjugacy" on γ up to and *including* b.

COROLLARY 4. *Let $\gamma(b)$ be the first conjugate point of $\gamma(a)$ along $\gamma : [a, b] \to M$. Then $\omega_n(\gamma) \leq i(\pi \circ \gamma)$, which is the total number of focal points $\gamma(t)$ of F_a on γ with $t < b$, and also the total number of conjugate points of $\pi\gamma(a)$ on $\pi \circ \gamma$ with $t < b$.*

In particular, conjugate points occur sooner in B: if $\gamma(a)$ has a conjugate point $\gamma(t)$, then there exists a conjugate point $\pi\gamma(t')$ of $\pi\gamma(a)$ with $t' \leq t$.

5. A special case. To illustrate preceding results we consider a submersion $\pi : M \to B$ for which M is compact and has constant positive curvature, say, $K = 1$. Examples are provided by the well-known Hopf fiberings of spheres over complex and quaternionic projective spaces [6], [3]. Let $\beta : R \to B$ be a unit-speed geodesic in B; we examine the distribution of the conjugate points of $\beta(0)$ along β. As mentioned in §2, there is a horizontal geodesic γ in M such that $\pi \circ \gamma = \beta$. Along γ the conjugate points of $\gamma(0)$ occur at $t = m\pi$, and have the maximum possible order, $\dim M - 1$. Hence in the notation of the previous section we have $\omega_\delta = \dim B - 1$, $\omega_n = k$ (fiber dimension), and $\omega_v = 0$ at each conjugate point. Thus Theorem 4 shows that (1) on β there are conjugate points of $\beta(0)$ at each $t = m\pi$, with order $\dim B - 1$. Theorem 5 and subsequent remarks show that (2) on an initial segment $\beta \mid [0, m\pi)$ there are conjugate points of total order at least mk. We call these *intermediate* conjugate points: they derive from strict focal points on γ, hence are distinct from the conjugate points in (1). The special character of conjugacy in M leads to a stronger version of (2): on each open interval $m\pi < t < (m + 1)\pi$ there are conjugate points $\beta(t)$ of $\beta(0)$ with total order at least k. To prove this, note that by Corollary 4 there are this many conjugate point *of* $\beta(m\pi)$ on the given interval; if $\beta(t)$ is one, then by Theorem 4, $\gamma(m\pi)$ is a focal point of F_t . The special nature of Jacobi fields on M is such that $\gamma(0)$ must also be a focal point of F_t , hence $\beta(0)$ and $\beta(t)$ are conjugate (orders the same throughout).

By an estimate on the curvature of B we can further restrict the location of the intermediate conjugate points. Because M has curvature 1, Corollary 1 of [6] gives this result: if x and y denote orthonormal tangent vectors to B—and also their horizontal lifts to M—then the sectional curvature of B on the plane determined by x and y is $1 + 3\ ||A_x y||^2$ (independent of choice of horizontal lift). Define the *norm* α *of* A to be the maximum of $||A_x y||$ for x, y as above. The norm $\alpha(\beta)$ of A along a geodesic β is similarly defined, but with the additional restriction that x be tangent to β. Thus the curvature of planes through β' is at most $1 + 3\ \alpha(\beta)^2$. It follows that on β the arc length between any pair of conjugate points is at most $\pi[1 + 3\alpha(\beta)^2]^{-\frac{1}{2}}$.

Now suppose that $\beta(t)$ is an intermediate conjugate point of $\beta(0)$, with $m\pi < t < (m + 1)\pi$. Again, Theorem 4 and the special character of Jacobi fields on M show that both $\gamma(m\pi)$ and $\gamma((m + 1)\pi)$ are focal points of F_t. Hence $\beta(t)$ is conjugate to both $\beta(m\pi)$ and $\beta((m + 1)\pi)$, so we conclude that $|t - m\pi|$ and $|(m + 1)\pi - t|$ are each at least $\pi[1 + 3\alpha(\beta)^2]^{-\frac{1}{2}}$. We summarize as follows:

COROLLARY 5. *Let* $\pi : M \to B$ *be a submersion for which* M *is compact and has constant sectional curvature* 1. *Let* $\beta : R \to B$ *be a unit-speed geodesic in* B, *and let* $\alpha(\beta)$ *be the norm of the tensor* A *along* β. *Then*

1. *For each integer* m, $\beta(m\pi)$ *is a conjugate point of* $\beta(0)$ *of order* $\dim B - 1$.
2. *All other conjugate points* $\beta(t)$ *of* $\beta(0)$ *lie in the intervals* $m\pi + d \leq t \leq (m + 1)\pi - d$, *where* $d = \pi[1 + 3\alpha(\beta)^2]^{-\frac{1}{2}}$.
3. *The total order of conjugacy on each such interval is at least the fiber dimension* k, *and each such intermediate conjugate point has order at most* k.

Because of (3) the intervals in (2) are non-empty if $k > 0$, hence $\alpha(\beta) \geq 1$ and $\alpha \geq 1$. We interpret this inequality roughly as follows. The definition of A shows that, along a horizontal geodesic, A measures the turbulence of the fibers; that is, the failure of the vertical distribution to be parallel. When the norm α of A is small, $\pi : M \to B$ is close to the Riemannian product case $B \times F \to B$, where $A = 0$. But the sectional curvature of the product $B \times F$ is frequently zero, so for $K_M = 1$, α cannot be too small.

For the sphere fibrations over CP^n and QP^n it is easy to see that α has the minimum value 1; these are submersions of spheres with minimum turbulence of fibers. In the context of the corollary above, if $\alpha = 1$, then for every geodesic β in B, $\alpha(\beta) \leq 1$; hence in each interval, as in (2), there is only one intermediate conjugate point—at $t = (m + \frac{1}{2})\pi$, with order k. Thus the geodesics of B have essentially the same conjugacy properties as for CP^n and QP^n, (where k is 1 and 3 respectively).

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